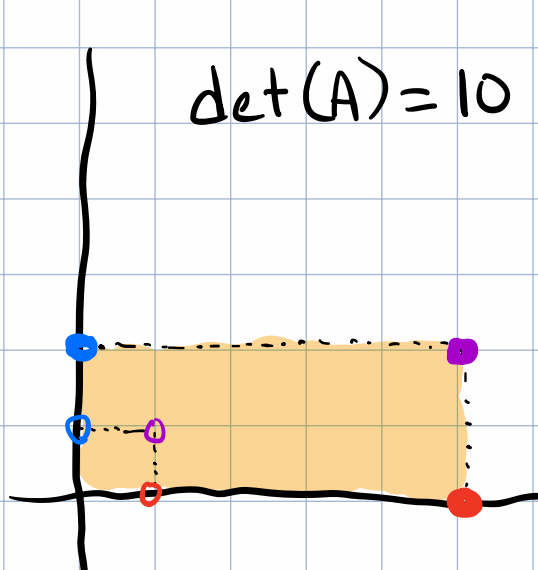


1) Given $\begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} = A$, Find $\det(A)$

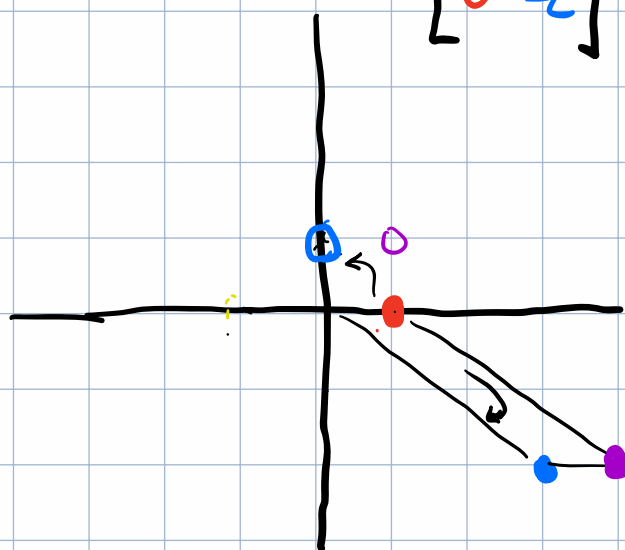


Is the determinant
is related to an
integral?

2) Does the transformation preserve or flip orientation?

$$\begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}$$

It flips orientation



3) Suppose a transformation doubles all lengths in every direction in 3D. What is its determinant?

Volume of a unit cube is $1 \cdot 1 \cdot 1 = 1$

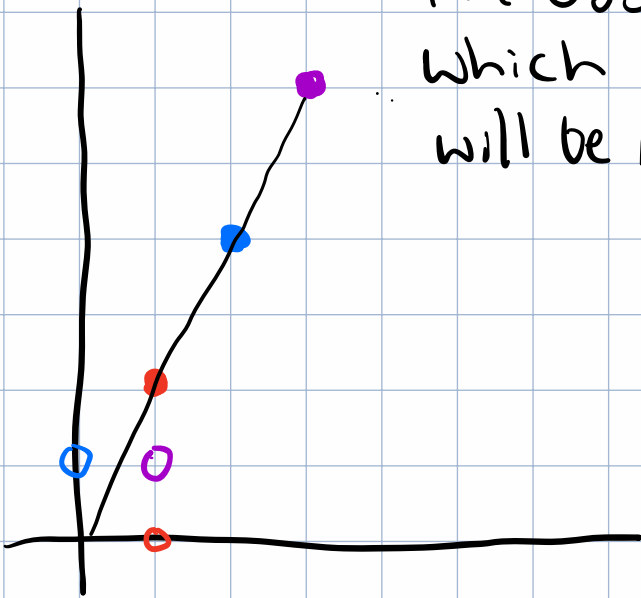
Volume of transformed cube is $2 \cdot 2 \cdot 2 = 8 = 2^3$

dimension
↓

4) Without calculating, is the following matrix invertable?

$$\begin{matrix} c_1 & c_2 \\ \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \end{matrix}$$

No. $c_2 = 2c_1$, therefore the determinant is 0, a matrix A is invertable iff $\det(A) \neq 0$



The basis vectors form a line, which has zero area, so all areas will be mapped to zero.